

Project: DSM MINI PROJECT 2026

Project Title: Self balancing and walking humanoid robot simulation , lower body joint analysis , torque analysis

Category: A

Team Members: Aryan hiware , daaris ahmed , hushan bhimani , jerome jose

Date: 24/04/2026

Q1: What was the single hardest technical challenge you faced in deriving the mathematical model? How did you resolve it?

The hardest technical challenge was accurately deriving the velocities and for each link in a multi-DOF, coupled system without making errors. Even small mistakes in kinematics would affect the entire Lagrangian and final equations. This was resolved by systematically defining the reference frames

Q2: Which Course Outcome (CO1-CO5) was most challenging for your team? Why?

The most challenging course outcome for this project is CO4: applying constitutive laws to analyze dynamic systems . This is because the humanoid robot is a multi-DOF system with nonlinear and coupled dynamics, making the Euler–Lagrange formulation complex. It requires accurate derivation of energies, velocities, and equations of motion, which is significantly more difficult than standard problems.

Q3: What would you do differently if you started the project over?

If I started the project again, I would begin with a simpler model, such as a reduced DOF system, and then gradually increase complexity. I would define reference frames and generalized coordinates more systematically at the start to avoid confusion later. I would plan the derivation in a more structured way, breaking it into smaller steps and validating each stage before proceeding.

Q4: What did you learn from this project that a regular textbook problem could NOT teach you?

I learned how to model a multi-DOF system where all joints are coupled, how small mistakes in kinematics or energy terms can completely affect the final equations, and how important it is to define reference frames and coordinates correctly from the start. It included dealing with assumptions, simplifications, and validation using simulations.

Q5: How did you divide work among team members? Did everyone understand the full derivation?

- Derivation & Modelling: aryan
- Simulation & Coding: daaris
- Validation & Analysis: hushan

- Presentation & Documentation: jerome
- All members discussed the derivation together, but not everyone initially understood every step. Through group discussions and cross-verification, the full model was eventually understood by all members.

Q6: What AI tools did you use? For what specific purpose?
What did you NOT use AI for?

AI tools (ChatGPT and Claude) were used for:

- Structuring derivations
- Debugging Python code
- Extracting physical and mathematical data from the urdf file

AI was not used for core derivation logic and writing and formatting of the report everything done on paper which was done manually based on course concepts.